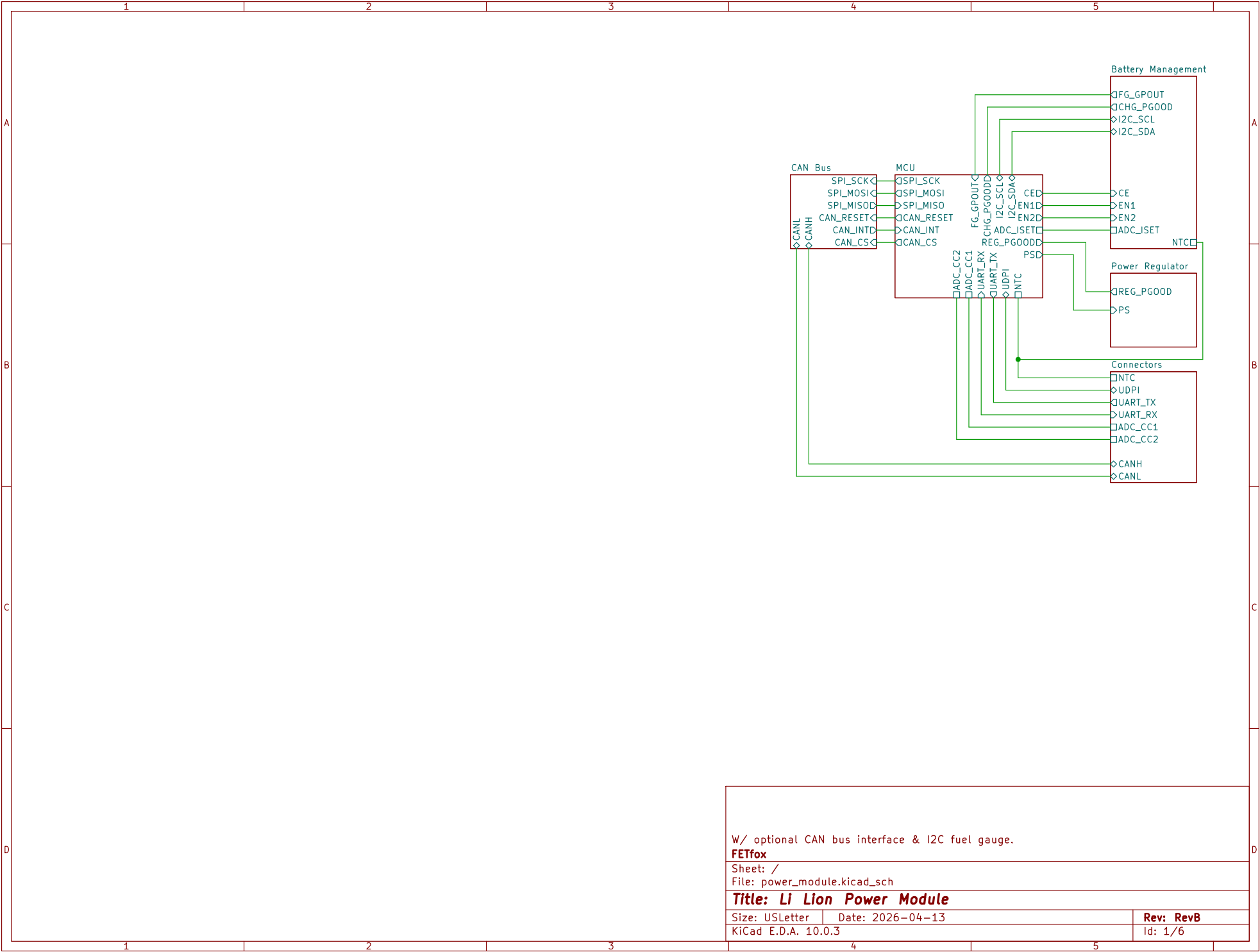




W/ optional CAN bus interface & I2C fuel gauge.

FETfox

Sheet: /

File: power_module.kicad_sch

Title: Li Lion Power Module

Size: USLetter Date: 2026-04-13

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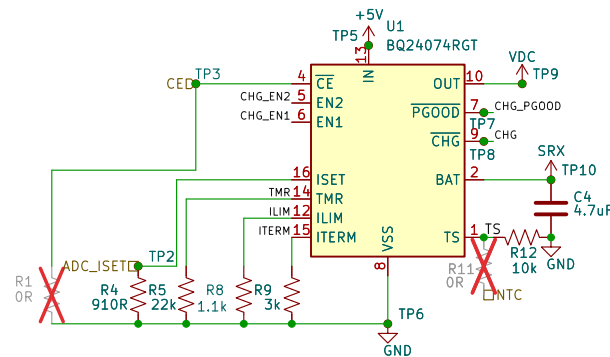
Rev: RevB

Id: 1/6

Li Ion Charger IC with Voltage and Current Protection IC with Optional I2C Fuel Gauge IC

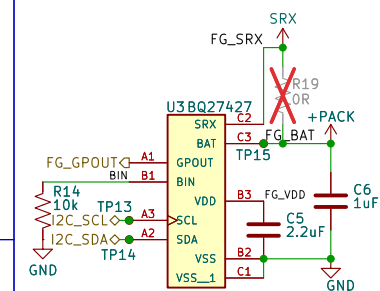
Populate R14 instead of R16 and R17 if host MCU is not present.

Q2 is a PMOS being used for polarity protection.



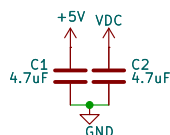
Li Ion Cell

R23 serves as a jumper for when the BQ27427 Fuel Gauge is not populated.

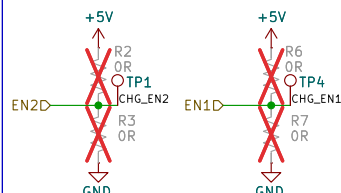


BQ27427 Fuel Gauge

Place decoupling capacitors as close to power pins as possible.

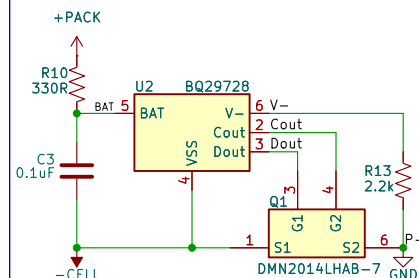


BQ24074 Decoupling Capacitors

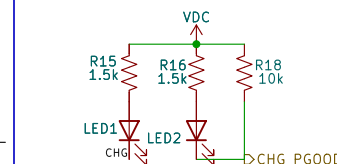


EN pin settings

EN2	EN1	INPUT CURRENT
0	0	100mA
0	1	500mA
1	0	Set by ILIM
1	1	Standby



BQ29728 Cell Protection Circuit



BQ24074 LED Pullups

W/ optional CAN bus interface & I2C fuel gauge.

FETfox

Sheet: /Battery Management/
File: batt_mgmt.kicad_sch

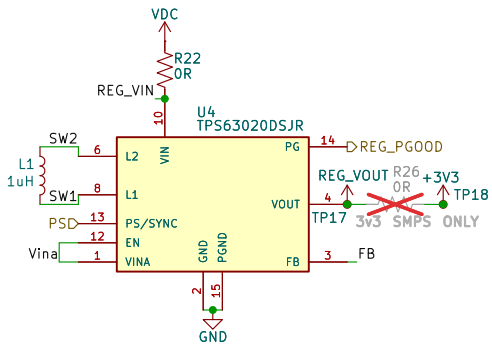
Title: Li Lion Power Module

Size: USLetter Date: 2026-04-13
KiCad E.D.A. 10.0.3

Rev: RevB
Id: 2/6

Boostbuck Regulator & Optional 3v3 LDO

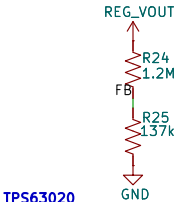
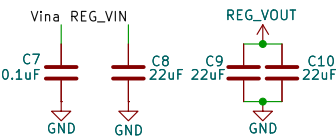
For 5v output, 3v3 LDO must be placed in series via populating R34.



TPS63020 Power Supply Regulator

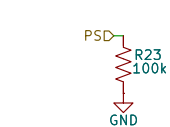
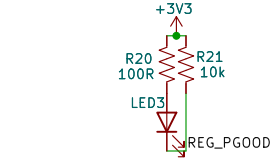
Place decoupling capacitors physically as close as possible to their corresponding pin.

Voltage output is set by R29: 768k resistor for 3v3 ouput 1.2M resistor for 5v



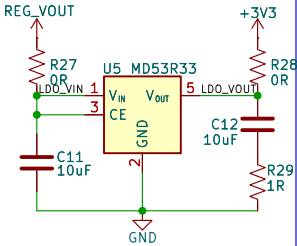
TPS63020 Decoupling Capacitors

TPS63020 Voltage feedback divider



TPS63020 PGOOD Pullup

TPS63020 PS Pulldown



MD53R33 3v3 LDO Regulator

W/ optional CAN bus interface & I2C fuel gauge.

FETfox

Sheet: /Power Regulator/

File: regulator.kicad_sch

Title: Li Lion Power Module

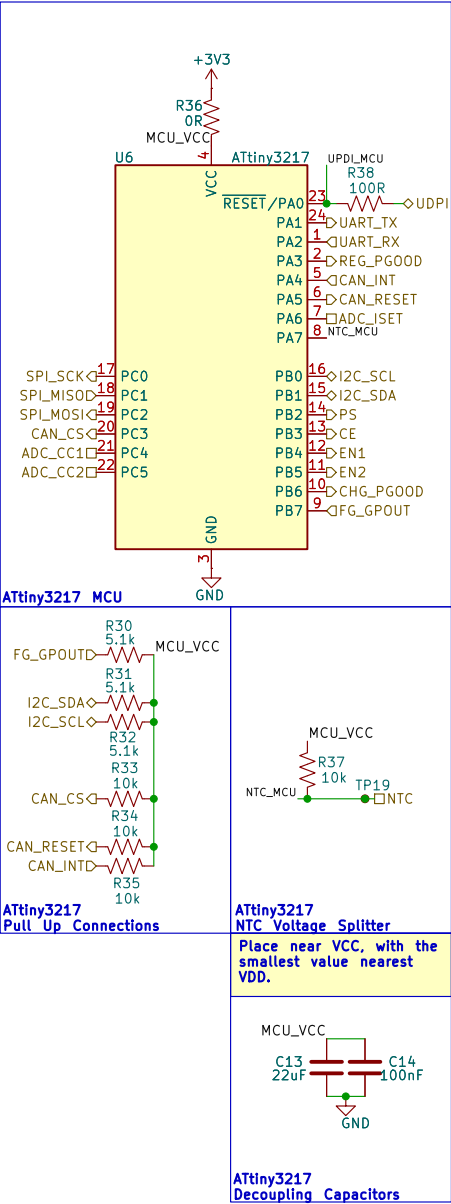
Size: USLetter Date: 2026-04-13

KiCad E.D.A. 10.0.3

Rev: RevB

Id: 3/6

Optional MCU



W/ optional CAN bus interface & I2C fuel gauge.

FETfox

Sheet: /MCU/

File: mcu.kicad_sch

Title: Li Lion Power Module

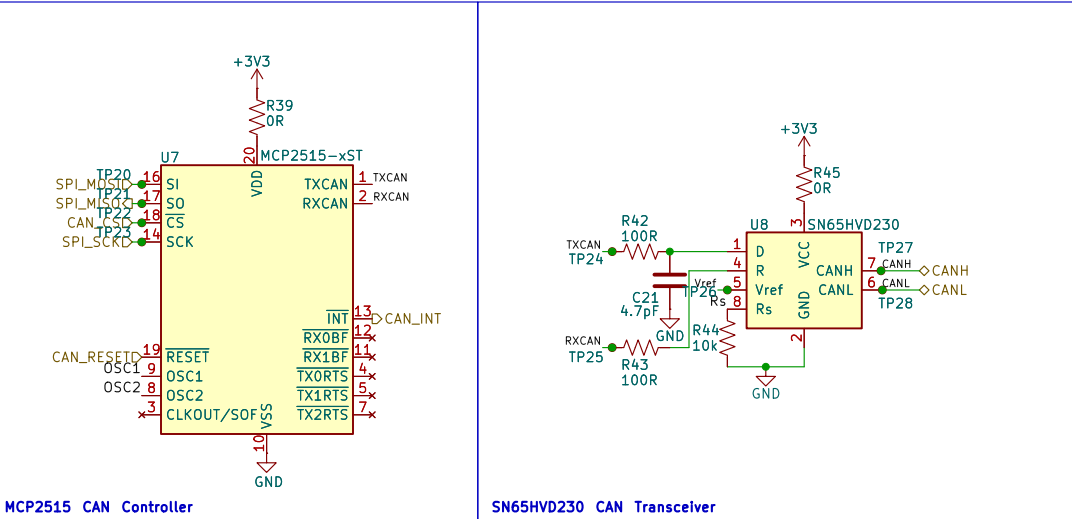
Size: USLetter Date: 2026-04-13

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Rev: RevB

Id: 4/6

Optional CAN Bus Interface



MCP2515 CAN Controller

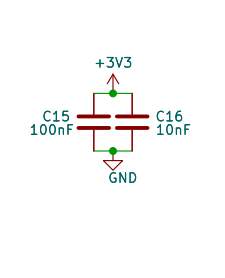
SN65HVD230 CAN Transceiver

Place near VDD, with the smallest value nearest VDD.

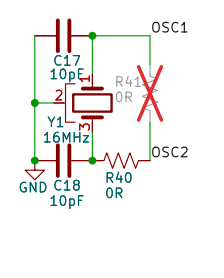
Place as close to OSC pins as possible.

Place near VCC, with the smallest value nearest VCC.

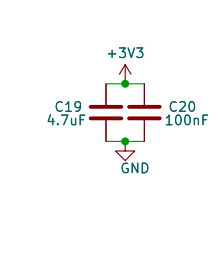
Place as close to SN65HVD230 as possible.



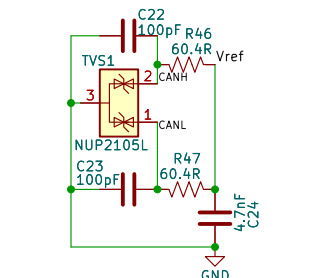
MCP2515 Decoupling Capacitors



MCP2515 Oscillator Circuit



SN65HVD230 Decoupling Capacitors



SN65HVD230 EMI Filtering & Termination

W/ optional CAN bus interface & I2C fuel gauge.

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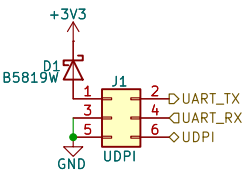
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File: CAN.kicad_sch

Title: Li Lion Power Module

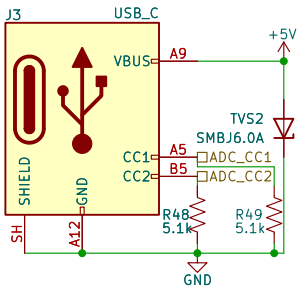
Size: USLetter Date: 2026-04-13
KiCad E.D.A. 10.0.3

Rev: RevB
Id: 5/6

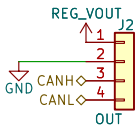
Connectors



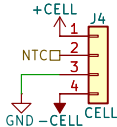
UPDI Connector



USB Connector



Output Connector



Pouch Cell / NTC Connector

W/ optional CAN bus interface & I2C fuel gauge.

FETfox

Sheet: /Connectors/
File: ext_conn.kicad_sch

Title: Li Lion Power Module

Size: USLetter Date: 2026-04-13
KiCad E.D.A. 10.0.3

Rev: RevB
Id: 6/6